BCG X • Executive Steering Committee Briefing

Project PowerCo: SME Customer Churn Prediction & Price Sensitivity Evaluation

1. SITUATION

Market Deregulation: European energy market liberalization has intensified competition across Small and Medium Enterprises (SMEs).

Baseline Churn Rate: PowerCo faces a baseline customer churn rate of **9.72%** (1,419 of 14,606 clients churned within 3 months).

Data Baseline: Audited 14,606 client profiles across 2015 alongside 193,002 monthly price data records across peak, mid-peak, and off-peak tariffs.

2. COMPLICATION

Client Hypothesis: PowerCo leadership hypothesized that price sensitivity is the primary driver of contract abandonment, proposing a blanket 20% tariff discount across accounts.

Commercial Risk: Granting blanket discounts would severely erode operating margins on loyal accounts (90.3% of client base) without guaranteeing churn prevention.

Data Imbalance: Severe 9:1 class imbalance masks true churn indicators under conventional accuracy metrics.

3. QUESTION & FINDINGS

Hypothesis Tested: Does price elasticity drive customer churn, and which metrics are most predictive?

Analytical Discovery: Price sensitivity is **not** the primary driver. Feature importance shows 12-month usage volume (**5.34%**) and net margin (**5.21%**) far outweigh price variations (**2.67%–2.98%**).

Driver Profile: High-margin, high-consumption clients churn due to overall margin burden and service tenure rather than small tariff swings.

4. ANSWER & STRATEGY

Model Recommendation: Deployed a 1,000-tree Random Forest classifier. By tuning probability threshold from 0.50 to **0.20**, **Recall increased from 4.6% to 35.5%** (catching 130 of 366 at-risk clients), tripling the F1-score.

Targeted Retention: Reject blanket price cuts. Deploy proactive retention discounts exclusively to top-decile risk accounts with high net margins.

Financial Impact: Preserves margin on 90% loyal customers while protecting high-value accounts at risk.